

Task 2: Write assembly language code to add 4 values directly in memory. [5]

- You can only use ax register
- Use direct addressing mod
- Also describe logical address of sum variable, like DS:0000

```
org 0x0100
```

```
mov ax, [num1]
mov [num1+10], ax
mov ax, [num1+2]
add [num1+10], ax
mov ax, [num1+4]
add [num1+10], ax
mov ax, [num1+6]
add [num1+10], ax
```

```
mov ax, 0x4c00
```

```
int 0x21
```

num1: dw 2,4,3,2,0 ; sum is stored at 5th index of array, it is stored at address **012A**

AX	0002	SI	0000	CS	19F5	IP	011B	Stack	+0 0000	Flags	7200
BX	0000	DI	0000	DS	19F5				+2 20CD		
CX	002A	BP	0000	ES	19F5	HS	19F5		+4 9FFF	OF DF IF SF ZF AF PF CF	
DX	0000	SP	FFFE	SS	19F5	FS	19F5		+6 EA00	0 0 1 0 0 0 0 0	

S or SI or SYM				1				A B C D E F 0 1			
CMD >S				DS:012A				0B 00 D8 01 C3 8B 07 8B			
0117 01062A01				ADD [012A],AX				DS:0132 57 02 85 D2 75 04 85 C0			
011B B8004C				MOV AX,4C00				DS:013A 74 1C C7 46 DC 00 00 8E			
011E CD21				INT 21				DS:0142 5E FC 83 7D 0E 00 74 09			
0120 0200				ADD AL,[BX+SI]				DS:014A 8B 46 F2 48 3B 46 F6 7E			
0122 0400				ADD AL,00				DS:0152 08 B8 01 00 EB 05 E9 42			
0124 0300				ADD AX,[BX+SI]				DS:015A 01 31 C0 89 46 E2 8B 46			
0126 0200				ADD AL,[BX+SI]				DS:0162 F6 D1 E0 D1 E0 C5 5E D8			
0128 0000				ADD [BX+SI],AL				DS:016A 01 C3 8B 37 8B 47 02 89			
012A 0B00				OR AX,[BX+SI]				DS:0172 46 FE 85 C0 75 04 85 F6			

2	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	
DS:0100	A1	20	01	A3	2A	01	A1	22	01	01	06	2A	01	A1	24	01	í .ú*.í" ...*.í\$.
DS:0110	01	06	2A	01	A1	26	01	01	06	2A	01	B8	00	4C	CD	21	..*.í&.. *.í.L=?
DS:0120	02	00	04	00	03	00	02	00	00	00	0B	00	D8	01	C3	8B	+.í
DS:0130	07	8B	57	02	85	D2	75	04	85	C0	74	1C	C7	46	DC	00	.íW.àµu. àLt.íF.
DS:0140	00	8E	5E	FC	83	7D	0E	00	74	09	8B	46	F2	48	3B	46	.Á^"à}... t.íF¿H:F

Task 3: Write assembly language code to add ten numbers using register + offset addressing. [5]

- Use register + offset addressing mod
- Use Loop.
- Define separate variable for sum and also write logical address.

```
org 0x0100
```

```
mov cx,10
mov ax,0
mov bx,0
```

```
l1:
    mov ax,[num+bx]
    add [sum],ax
    add bx,2
    dec cx
    jnz l1
```

```
mov ax,0x4c00
int 0x21
```

```
num: dw 1,2,3,4,5,6,7,8,9,0
```

```
sum: dw 0 ; sum is stored at address 0131
```

Name: dwc										; earnings stored at address 0101																			
AX	4C00	SI	0000	CS	19F5	IP	011B	Stack	+0 0000	Flags	7244																		
BX	0014	DI	0000	DS	19F5					+2 20CD																			
CX	0000	BP	0000	ES	19F5	HS	19F5					+4 9FFF	OF	DF	IF	SF	ZF	AF	PF	CF									
DX	0000	SP	FFFE	SS	19F5	FS	19F5					+6 EA00	0	0	1	0	1	0	1	0									
S or SI or SYM																													
CMD >S										1										0 1 2 3 4 5 6 7									
0118 B8004C										MOV AX,4C00										DS:0000 CD 20 FF 9F 00 EA F0 FE									
011B CD21										INT 21										DS:0008 AD DE 1B 05 C5 06 00 00									
011D 0100										ADD [BX+SI],AX										DS:0010 18 01 10 01 18 01 92 01									
011F 0200										ADD AL,[BX+SI]										DS:0018 01 01 01 00 02 FF FF FF									
0121 0300										ADD AX,[BX+SI]										DS:0020 FF FF FF FF FF FF FF									
0123 0400										ADD AL,00										DS:0028 FF FF FF FF EB 19 C0 11									
0125 050006										ADD AX,0600										DS:0030 A2 01 14 00 18 00 F5 19									
0128 0007										ADD [BX],AL										DS:0038 FF FF FF FF 00 00 00 00									
012A 0008										ADD [BX+SI],CL										DS:0040 05 00 00 00 00 00 00									
																				DS:0048 00 00 00 00 00 00 00									
2																													
DS:0131		2D	00	02	85	D2	75	04	85	C0	74	1C	C7	46	DC	00	00	-..àµu.à Lt. F■..											
DS:0141		8E	5E	FC	83	7D	0E	00	74	09	8B	46	F2	48	3B	46	F6	â"â¿.t .iF¿H;F÷											
DS:0151		7E	08	B8	01	00	EB	05	E9	42	01	31	C0	89	46	E2	8B	~.j..δ.θ B.1¿FFi											
DS:0161		46	F6	D1	E0	D1	E0	C5	5E	D8	01	C3	8B	37	8B	47	02	F÷TαTα÷ †. †i7iG.											
DS:0171		89	46	FE	85	C0	75	04	85	F6	74	12	8E	D8	83	7C	0C	ēF■àLu.à ÷t.†¿†ài											

AX	0001	SI	000A	CS	19F5	IP	0114	Stack	+0	0000	Flags	7200									
BX	0000	DI	0000	DS	19F5				+2	20CD											
CX	0001	BP	0000	ES	19F5	HS	19F5		+4	9FFF	OF	DF IF SF ZF AF PF CF									
DX	0000	SP	FFFE	SS	19F5	FS	19F5		+6	EA00	0	0 1 0 0 0 0 0									
S or SI or SYM																					
CMD >S								1					0 1 2 3 4 5 6 7								
0110	89843001			MOV	[0130+SI],AX			DS:0000	CD	20 FF 9F 00 EA F0 FE											
0114	81C60200			ADD	SI,0002			DS:0008	AD DE 1B 05 C5 06 00 00												
0118	81EB0200			SUB	BX,0002			DS:0010	18 01 10 01 18 01 92 01												
011C	49			DEC	CX			DS:0018	01 01 01 00 02 FF FF FF												
011D	75ED			JNZ	010C			DS:0020	FF FF FF FF FF FF FF FF												
011F	B8004C			MOV	AX,4C00			DS:0028	FF FF FF FF EB 19 C0 11												
0122	CD21			INT	21			DS:0030	A2 01 14 00 18 00 F5 19												
0124	0100			ADD	[BX+SI],AX			DS:0038	FF FF FF FF 00 00 00 00												
0126	0200			ADD	AL,[BX+SI]			DS:0040	05 00 00 00 00 00 00 00												
								DS:0048	00 00 00 00 00 00 00 00												
2													0 1 2 3 4 5 6 7	8 9 A B C D E F							
DS:0130	06 00 05 00 04 00 03 00	02 00 01 00 C7 46 DC 00									 F .										
DS:0140	00 8E 5E FC 83 7D 0E 00	74 09 8B 46 F2 48 3B 46									..^~a).. t.iF2H:F										
DS:0150	F6 7E 08 BB 01 00 EB 05	E9 42 01 31 C0 89 46 E2									~.~.~.δ. 0B.1eFR										
DS:0160	8B 46 F6 D1 E0 D1 E0 C5	5E D8 01 C3 8B 37 8B 47									iF=7x~t ^+.i7ig										
DS:0170	02 89 46 FE 85 C0 75 04	85 F6 74 12 8E D8 83 7C									.eFa~u. à+t.Afâ!										
1 Step													2ProcStep	3Retrieve	4Help ON	5BRK Menu	6	7 up	8 dn	9 le	10 ri